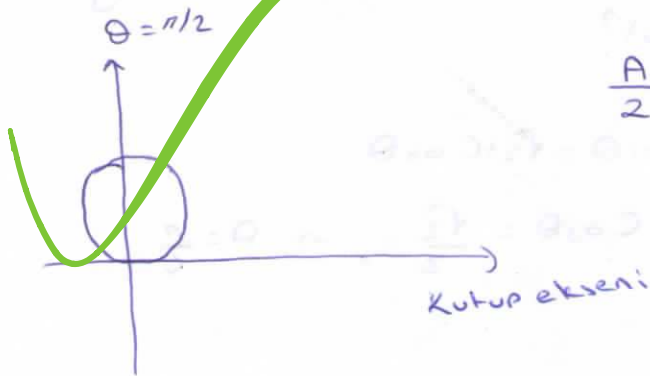


* $r = \sqrt{2} \sin \theta$ ile sınırlı bölgenin alanı?



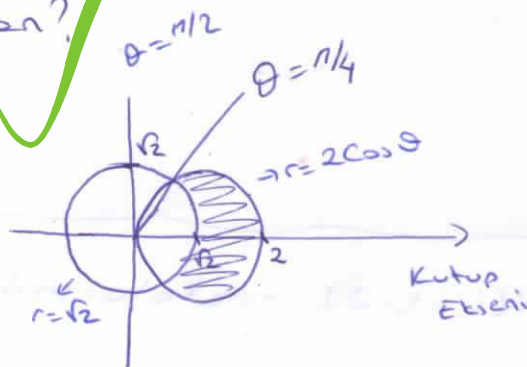
$$\frac{A}{2} = \frac{1}{2} \int_0^{\pi/2} (\sqrt{2} \sin \theta)^2 d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} 2 \sin^2 \theta d\theta$$

$$= \int_0^{\pi/2} \frac{1 - \cos 2\theta}{2} d\theta = \frac{\pi}{4}$$

$$\boxed{A = \frac{\pi}{2}}$$

* a) $r = 2 \cos \theta$ eğrisinin içinde $r = \sqrt{2}$ nin dışında kalan alan?



$$2 \cos \theta = \sqrt{2} \rightarrow \boxed{\theta = \frac{\pi}{4}}$$

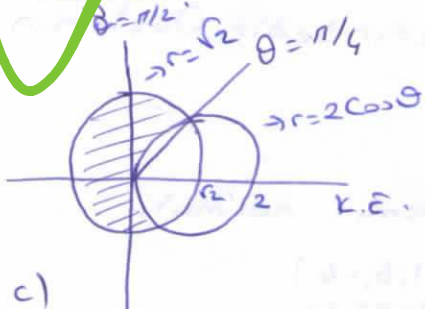
$$\frac{A}{2} = \frac{1}{2} \int_0^{\pi/4} (2 \cos \theta)^2 d\theta - \frac{1}{2} \int_0^{\pi/4} (\sqrt{2})^2 d\theta$$

$$= \frac{1}{2} \left[\int_0^{\pi/4} (4 \cos^2 \theta - 2) d\theta \right]$$

$$= \frac{1}{2} \int_0^{\pi/4} 2 \cos 2\theta d\theta = \frac{\sin 2\theta}{2} \Big|_0^{\pi/4} = \frac{1}{2}$$

$$\boxed{A = 1}$$

b) $r = 2 \cos \theta$ nin dışında, $r = \sqrt{2}$ nin içinde kalan alanı veren integral:



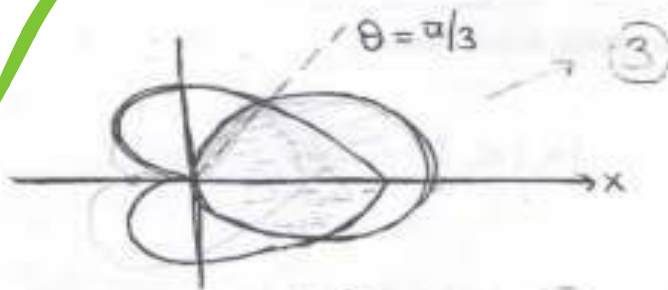
$$\frac{A}{2} = \frac{1}{2} \int_{\pi/4}^{\pi/2} (\sqrt{2})^2 d\theta - \frac{1}{2} \int_{\pi/4}^{\pi/2} (2 \cos \theta)^2 d\theta$$

c) Ortak Alanı veren integral:



$$\frac{A}{2} = \frac{1}{2} \int_0^{\pi/4} (\sqrt{2})^2 d\theta + \frac{1}{2} \int_{\pi/4}^{\pi/2} (2 \cos \theta)^2 d\theta$$

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4. a) $r = 3\cos\theta$ ve $r = 1 + \cos\theta$ eğrilerinin içinde kalan bölgenin alanını veren integrali yazınız.
(Integral hesaplanmayacak.)



a)

$$\frac{A}{2} = \frac{1}{2} \int_0^{\pi/3} (1 + \cos\theta)^2 d\theta + \frac{1}{2} \int_{\pi/3}^{\pi/2} (3\cos\theta)^2 d\theta$$

b) Kardioid içi, çember dışı

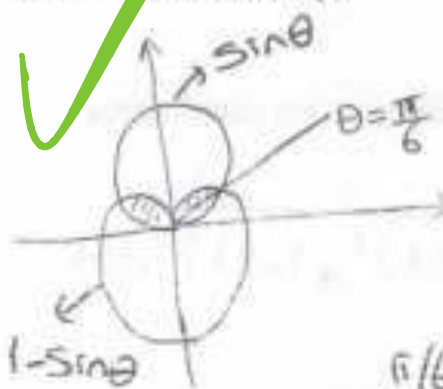
$$\frac{A}{2} = \frac{1}{2} \int_{\pi/3}^{\pi} (1 + \cos\theta)^2 d\theta - \frac{1}{2} \int_{\pi/3}^{\pi/2} (3\cos\theta)^2 d\theta$$

c) Çember içi, kardioid dışı

$$\frac{A}{2} = \frac{1}{2} \int_0^{\pi/3} (3\cos\theta)^2 - (1 + \cos\theta)^2 d\theta$$

b) $\rho = 1 - \sin\theta$ kardioidi ve $\rho = \sin\theta$ çemberinin her ikisinin de içinde kalan bölgenin alanını bulunuz. (12p)

$$1 - \sin\theta = \sin\theta \Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

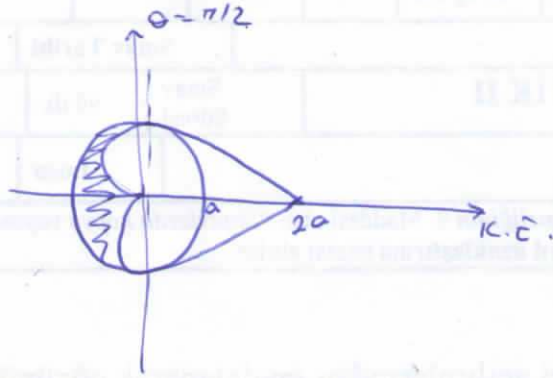


$$\frac{A}{2} = \frac{1}{2} \int_0^{\pi/6} (\sin\theta)^2 d\theta + \frac{1}{2} \int_{\pi/6}^{\pi/2} (1 - \sin\theta)^2 d\theta$$

$$A = \int_0^{\pi/6} \frac{1 - \cos 2\theta}{2} d\theta + \int_{\pi/6}^{\pi/2} \frac{1}{2} (3 - 4\sin\theta - \cos 2\theta) d\theta$$

$$A = \left(\frac{\pi}{12} - \frac{\sqrt{3}}{8} \right) + \left(\frac{\pi}{2} - \frac{7\sqrt{3}}{8} \right) = \frac{7\pi}{12} - \sqrt{3} \text{ br}^2$$

2) $a > 0$ olmak üzere $r = a(1 + \cos \theta)$ kardioidinin dışında, $r = a$ çemberinin içinde kalan bölgenin alanını hesaplayınız. (Şekil çiziniz)



$$\frac{A}{2} = \int_{\pi/2}^{\pi} a^2 - (a + a \cos \theta)^2 d\theta = \int_{\pi/2}^{\pi} (-2a^2 \cos \theta - a^2 \underbrace{\cos^2 \theta}_{\frac{1 + \cos 2\theta}{2}}) d\theta$$

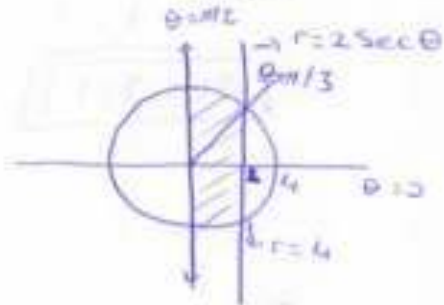
$$= -2a^2 \sin \theta - a^2 \frac{\theta}{2} - \frac{a^2}{4} \sin 2\theta \Big|_{\pi/2}^{\pi}$$

$$= -a^2 \frac{\pi}{2} - \left(-2a^2 - a^2 \frac{\pi}{4} \right) = -\frac{a^2 \pi}{2} + 2a^2 + \frac{a^2 \pi}{4}$$

$$= 2a^2 - \frac{\pi}{4} a^2$$

3) $r = 1$, $\theta = \frac{\pi}{2}$, $r = 2 \sec \theta$ arasında kalan bölgenin alanını integral ile bulunuz.

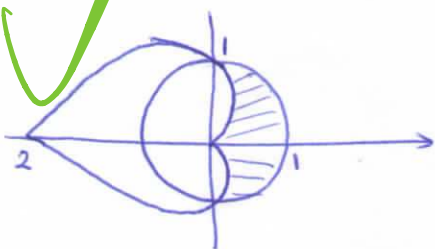
$$r = 2 \sec \theta = \frac{2}{\cos \theta} \Rightarrow r \cos \theta = 2 \Rightarrow \boxed{x=2} \text{ doğrusu}$$



$$2 \sec \theta = 4 \Rightarrow \sec \theta = 2 \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = \pi/3$$

$$\frac{A}{2} = \frac{1}{2} \int_0^{\pi/3} (2 \sec \theta)^2 d\theta + \frac{1}{2} \int_{\pi/3}^{\pi/2} (1)^2 d\theta$$

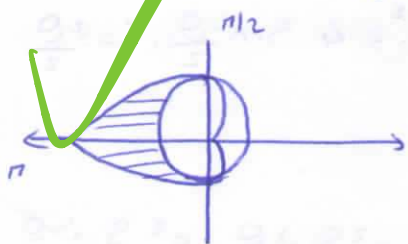
* $r=1$ çemberinin içinde, $r=1-\cos\theta$ kardioidinin dışında kalan bölgenin alanını veren integral?



$$\frac{A}{2} = \int_0^{\pi/2} \frac{1}{2} \cdot d\theta - \frac{1}{2} \int_0^{\pi/2} (1-\cos\theta)^2 d\theta$$

$$A = \int_0^{\pi/2} (1 - (1-\cos\theta)^2) d\theta$$

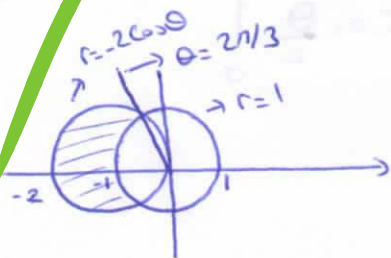
b) çemberin dışı, kardioidin içi:



$$\frac{A}{2} = \int_{\pi/2}^{\pi} \frac{1}{2} \cdot (1-\cos\theta)^2 d\theta - \int_{\pi/2}^{\pi} \frac{1}{2} d\theta$$

$$A = \int_{\pi/2}^{\pi} ((1-\cos\theta)^2 - 1) d\theta$$

* $r=-2\cos\theta$ çemberinin içinde, $r=1$ çemberinin dışında kalan alan?



$$-2\cos\theta = 1 \Rightarrow \theta = \frac{2\pi}{3}$$

$$\frac{A}{2} = \frac{1}{2} \int_{2\pi/3}^{\pi} (1-2\cos\theta)^2 d\theta - \frac{1}{2} \int_{2\pi/3}^{\pi} 1^2 d\theta$$

$$A = \int_{2\pi/3}^{\pi} \left(4\cos^2\theta - 4\cos\theta + 1\right) d\theta = \int_{2\pi/3}^{\pi} (1 + 2\cos 2\theta) d\theta$$

$$= \theta + \sin 2\theta \Big|_{2\pi/3}^{\pi} = \pi - \frac{2\pi}{3} - \sin \frac{\pi}{3} = \frac{\pi}{3} - \frac{\sqrt{3}}{2}$$

* $r = a \sin^2 \frac{\theta}{2}$ eğrisinin $0 \leq \theta \leq \pi$ aralığındaki uzunluğu? ($a > 0$)

$$S = \int_0^{\pi} \sqrt{r^2 + (r')^2} d\theta$$

$$r^2 = a^2 \sin^4 \frac{\theta}{2}$$

$$r' = a \cdot 2 \cdot \sin \frac{\theta}{2} \cdot \cos \frac{\theta}{2} \cdot \frac{1}{2} = a \cdot \sin \frac{\theta}{2} \cdot \cos \frac{\theta}{2}$$

$$(r')^2 = a^2 \sin^2 \frac{\theta}{2} \cdot \cos^2 \frac{\theta}{2}$$

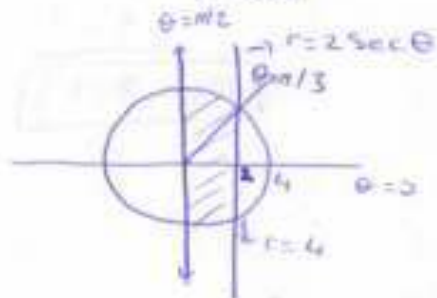
$$r^2 + (r')^2 = a^2 \sin^4 \frac{\theta}{2} + a^2 \sin^2 \frac{\theta}{2} \cdot \overbrace{\cos^2 \frac{\theta}{2}}^{1 - \sin^2 \frac{\theta}{2}} = a^2 \cancel{\sin^4 \frac{\theta}{2}} + a^2 \sin^2 \frac{\theta}{2} - a^2 \cancel{\sin^4 \frac{\theta}{2}} = a^2 \sin^2 \frac{\theta}{2}$$

$$\sqrt{r^2 + (r')^2} = \sqrt{a^2 \sin^2 \frac{\theta}{2}} = a \cdot \left| \sin \frac{\theta}{2} \right|$$

$$S = \int_0^{\pi} a \cdot \left| \sin \frac{\theta}{2} \right| d\theta = \int_0^{\pi} a \cdot \sin \frac{\theta}{2} d\theta = -2a \cos \frac{\theta}{2} \Big|_0^{\pi} = \boxed{2a}$$

* $r = 4$, $\theta = \frac{\pi}{2}$, $r = 2 \sec \theta$ arasında kalan bölgenin alanı nedir? integral?

$$r = 2 \sec \theta = \frac{2}{\cos \theta} \Rightarrow r \cos \theta = 2 \Rightarrow \boxed{x=2} \text{ doğrusu}$$



$$2 \sec \theta = 4 \Rightarrow \sec \theta = 2 \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

$$\frac{A}{2} = \frac{1}{2} \int_0^{\pi/3} (2 \sec \theta)^2 d\theta + \frac{1}{2} \int_{\pi/3}^{\pi/2} (4)^2 d\theta$$

⊗ a) $r = 3\sin\theta$, $r = 1 + \sin\theta$ ortak alan?

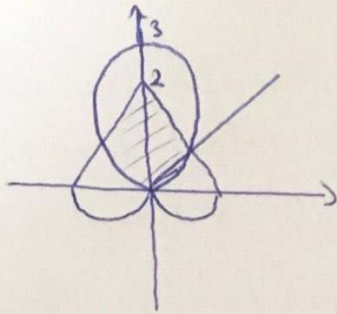
b) $r = 3\sin\theta$ içi, $r = 1 + \sin\theta$ dışı alan?

c) $r = 3\sin\theta$ dışı, $r = 1 + \sin\theta$ içi alan?

$$3\sin\theta = 1 + \sin\theta \Rightarrow \sin\theta = \frac{1}{2}$$

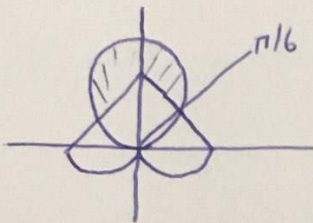
$$\theta = \frac{\pi}{6}$$

a)



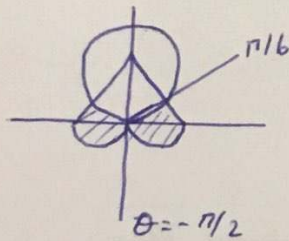
$$\frac{A}{2} = \frac{1}{2} \int_0^{\pi/6} (3\sin\theta)^2 d\theta + \frac{1}{2} \int_{\pi/6}^{\pi/2} (1 + \sin\theta)^2 d\theta$$

b)



$$\frac{A}{2} = \frac{1}{2} \int_{\pi/6}^{\pi/2} (3\sin\theta)^2 - (1 + \sin\theta)^2 d\theta$$

c)

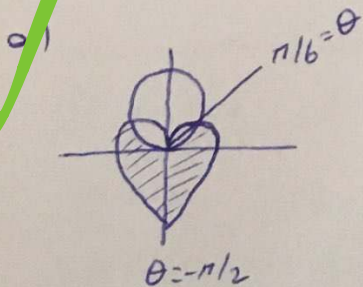


$$\frac{A}{2} = \frac{1}{2} \int_{-\pi/2}^{\pi/6} (1 + \sin\theta)^2 d\theta - \frac{1}{2} \int_0^{\pi/6} (3\sin\theta)^2 d\theta$$

⊗ a) $r = 1 - \sin\theta$ içi $r = \sin\theta$ dışı alan?

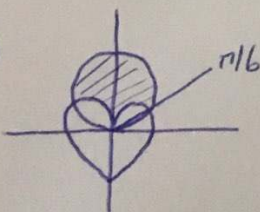
b) $r = 1 - \sin\theta$ dışı $r = \sin\theta$ içi alan? $1 - \sin\theta = \sin\theta$

$$\sin\theta = \frac{1}{2} \quad \theta = \frac{\pi}{6}$$



$$\frac{A}{2} = \frac{1}{2} \int_{-\pi/2}^{\pi/6} (1 - \sin\theta)^2 d\theta - \frac{1}{2} \int_0^{\pi/6} (\sin\theta)^2 d\theta$$

b)



$$\frac{A}{2} = \frac{1}{2} \int_{\pi/6}^{\pi/2} ((\sin\theta)^2 - (1 - \sin\theta)^2) d\theta$$